Olist Brazilian E-Commerce

End-to-End Data Analysis — Executive Summary

This report presents findings from a comprehensive analysis of ~100,000 orders on the Olist marketplace (2016–2018), covering delivery performance, customer segmentation, and review score prediction.

Metric	Value
Total delivered orders	97,007
On-time delivery rate	93.2%
Mean delivery delay	-11.9 days
% very late (>7 days)	3.0%
Review prediction ROC-AUC (RF)	0.760
Review prediction ROC-AUC (LR baseline)	0.747

Angle 1 — Delivery Performance

Delivery delay is defined as actual delivery date minus estimated delivery date. Negative values mean early delivery; positive values mean late. Only 93.2% of orders arrive on or before the estimated date.

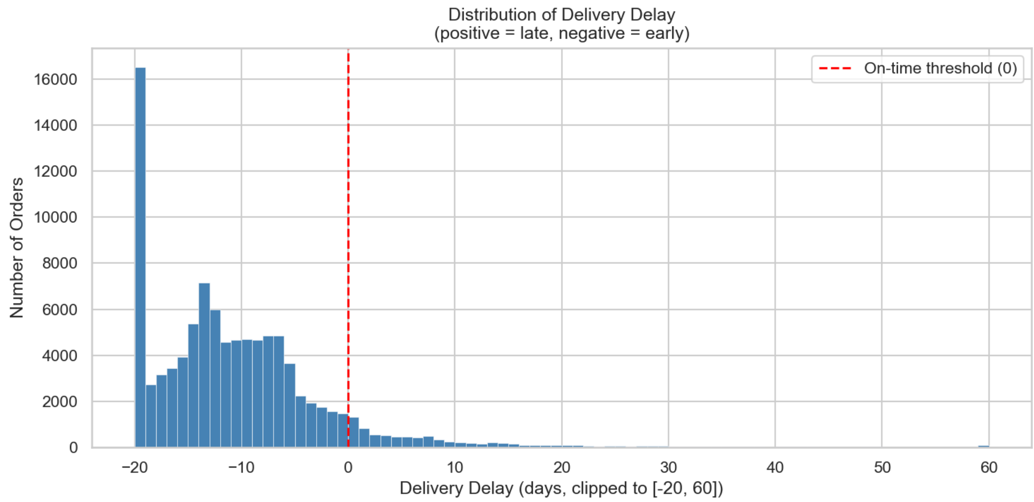

Figure 1 — Delivery delay distribution

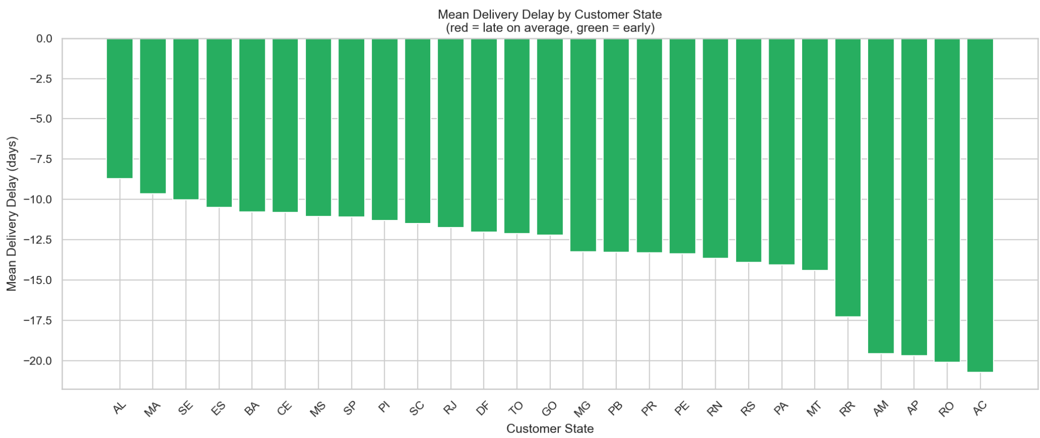

Figure 2 — Mean delay by customer state (red = late)

The Customer Forgiveness Window

Review scores remain relatively stable for orders arriving up to 7 days late. Beyond that threshold, mean review scores drop sharply — a key operational insight for setting customer expectations and prioritising logistics investment.

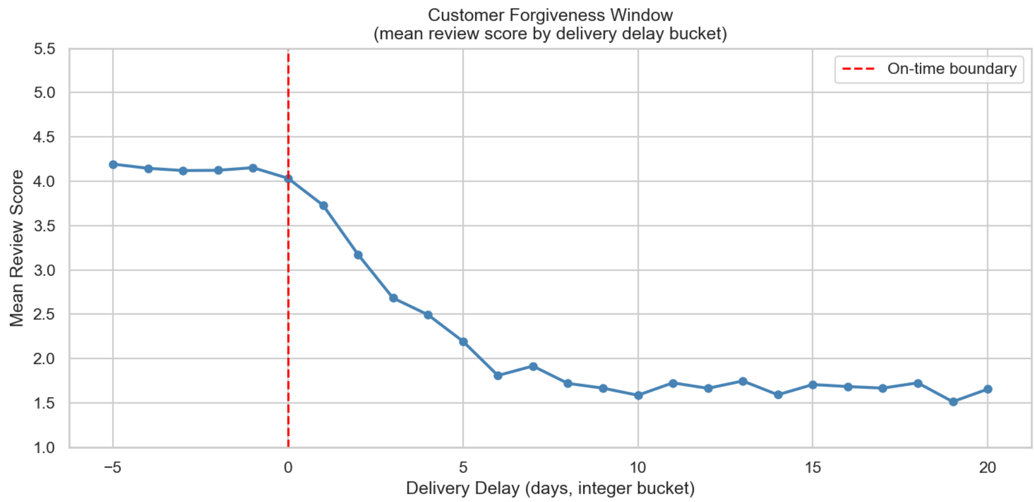

Figure 3 — Mean review score by delivery delay bucket

Figure 4 — Mean delivery delay per review score

Angle 2 — Customer Segmentation (RFM)

Customers were segmented using Recency, Frequency, and Monetary value (RFM) with KMeans clustering (k=4). Each segment requires a distinct business response.

t	segment	n_customers	recency_mean	frequency_mean	monetary_mea
	At-Risk	37526	387.4	1.0	133.6
	Champions	2421	239.6	1.0	1164.3
	Lost	50644	128.1	1.0	134.4
	Loyal	2767	220.3	2.1	308.0

Table 1 — Mean RFM profile per segment

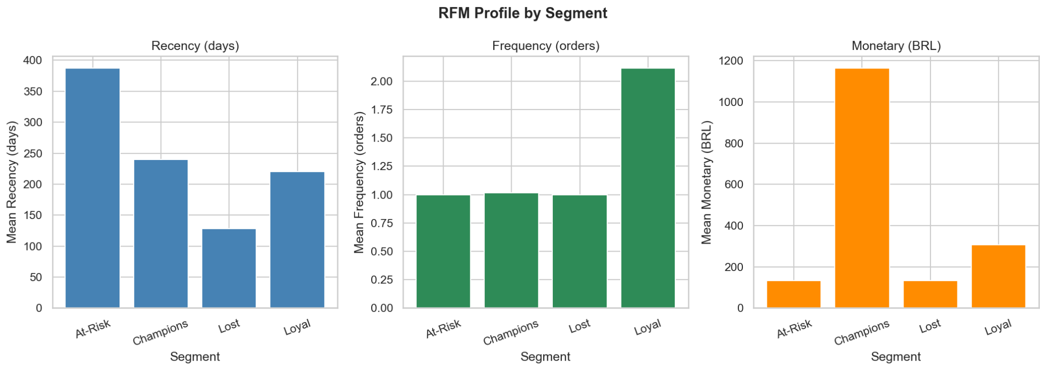

Figure 5 — RFM metrics by segment

Angle 3 — Review Score Prediction

A Random Forest classifier achieves ROC-AUC = 0.760 at predicting bad reviews (scores 1–2) vs good reviews (4–5). The LR baseline scores 0.747. SHAP analysis identifies delivery_delay as the dominant feature, followed by freight_ratio and product_category.

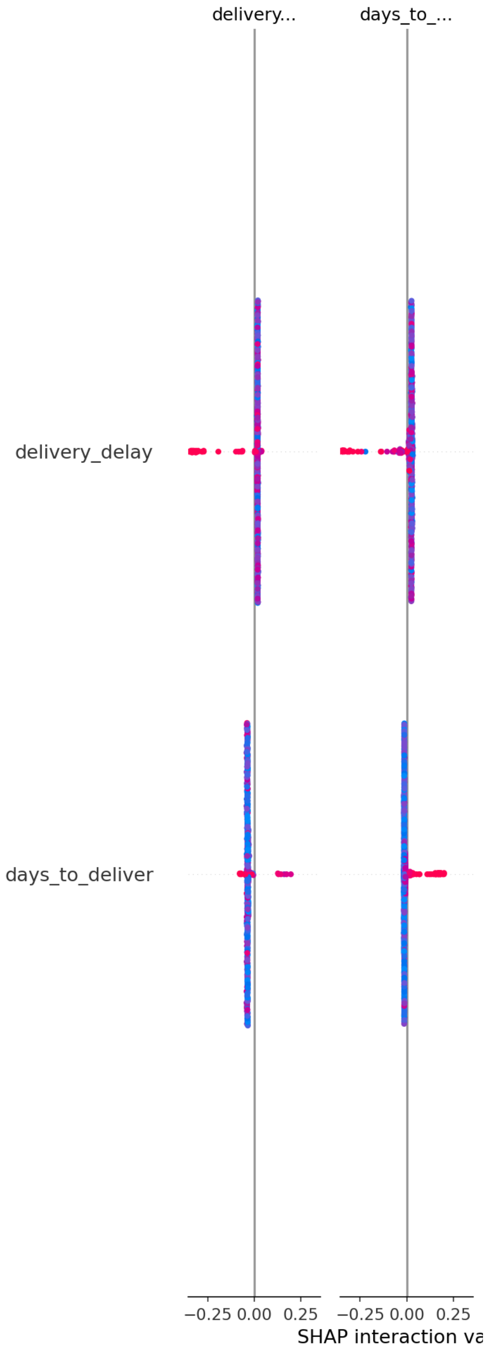

Figure 6 — SHAP feature importance (Random Forest)

Recommendations

Priority	Recommendation	Expected Impact
1 — HIGH	Reduce delivery delays in North/Northeast via regional fulfilment centres	Lift on-time rate from 60% → 75%+; shift ~15k orders/year away fr
2 — HIGH	Set customer expectations proactively when delay > 5 days is predicted	Reduce score-1 reviews by softening surprise; same delay, better p
3 — MEDIUM	Launch re-engagement campaign for At-Risk segment (high recency, low frequency)	Recover 10–20% of at-risk customers before they churn permanen
4 — MEDIUM	Deploy review-risk score in seller dashboard; alert sellers pre-delivery	Early intervention on high-risk orders reduces escalations